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A Bayesian Dirichlet auto-regressive moving average model for forecasting lead times

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ABSTRACT

In the hospitality industry, lead time data are a form of compositional data that are crucial for business planning, resource allocation, and staffing. Hospitality businesses accrue fees daily, but recognition of these fees is often deferred. This paper presents a novel class of Bayesian time series models, the Bayesian Dirichlet auto-regressive moving average (B-DARMA) model, designed specifically for compositional time series. The model is motivated by the analysis of five years of daily fees data from Airbnb, with the aim of forecasting the proportion of future fees that will be recognized in 12 consecutive monthly intervals. Each day's compositional data are modeled as Dirichlet distributed, given the mean and a scale parameter. The mean is modeled using a vector auto-regressive moving average process, which depends on previous compositional data, previous compositional parameters, and daily covariates. The B-DARMA model provides a robust solution for analyzing large compositional vectors and time series of varying lengths. It offers efficiency gains through the choice of priors, yields interpretable parameters for inference, and produces reasonable forecasts. The paper also explores the use of normal and horseshoe priors for the vector auto-regressive and vector moving average coefficients, and for regression coefficients. The efficacy of the B-DARMA model is demonstrated through simulation studies and an analysis of Airbnb data.

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1. Introduction

Travel, entertainment, and hospitality businesses earn fees each day, but these fees cannot be recognized until later. The *lead time* is the amount of time until fees earned on a given day can be recognized. Future dates when fees may be recognized are allocated into regular intervals, forming a partition of the future, often weekly, monthly, or annual time intervals. Each day, another vector of fee allocations is observed. The distribution of fees into future intervals can be analyzed separately from the total amount of fees; fractional allocations into future intervals sum to one and form a compositional time series. The basic compositional observation is a continuous vector of probabilities that a dollar earned today can be recognized in each future interval. We wish to understand the process generating the compositional time series and forecast the series into the future. This information can be used by businesses for the allocation of resources, for business planning, and for staffing.

We analyze five years of daily fees billed at Airbnb that will be recognized in the future. Future fees are allocated to one of 12 consecutive monthly intervals; fees beyond this range are small and ignored in this analysis. More granular inference on the lead time of revenue recognition

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would not improve business planning or resource allocation, though it substantially complicates modeling, model fitting, and the communication of results. To forecast a current day's allocations we have all prior days' data and we have deterministic characteristics of days such as day of the week, season, and sequential day of the year.

A lead time compositional time series $\mathbf{y}_t = (y_{t1}, \ldots, y_{tJ})'$ is a multivariate J-vector time series with observed data y_{tj}, where $t = 1, \ldots, T$ indexes consecutive days or other time units, $j = 1, \ldots, J$ indexes the J future revenue recognition intervals, and $0 < y_{tj} < 1$ and $\sum_{j=1}^{J} y_{tj} = 1$ for all t. A natural model for compositional data is the Dirichlet distribution, which is in the exponential family of distributions, and thus Dirichlet time series are special cases of generalized linear time series. Benjamin, Rigby, and Stasinopoulos (2003) proposed a univariate generalized ARMA data model in a frequentist framework.

There are numerous books on Bayesian analysis of time series data (Barber, Cemgil, & Chiappa, 2011; Berliner, 1996; Koop & Korobilis, 2010; Pole, West, & Harrison, 2018; Prado & West, 2010; West, 1996) as well as papers on Bayesian vector auto-regressive (AR) (VAR) moving average (MA) (VMA) (ARMA)(VARMA) time series models (Bańbura, Giannone, & Reichlin, 2010; Karlsson, 2013; Spencer, 1993; Uhlig, 1997), and on Bayesian generalized linear time series models (Silveira de Andrade, Andrade, & Ehlers, 2015; Berry & West, 2020; Brandt & Sandler, 2012; Chen & Lee, 2016; Fukumoto, Beger, & Moore, 2019; McCabe & Martin, 2005; Nariswari & Pudjihastuti, 2019, 2019; Roberts & Penny, 2002; West, 2013).

Dirichlet time series data are less commonly modeled in the literature. Grunwald, Raftery, and Guttorp (1993) proposed a Bayesian compositional state space model with data modeled as a Dirichlet distribution given a mean vector, and with the current mean given the prior mean also modeled as a Dirichlet distribution. They did not use Markov chain Monte Carlo (MCMC) for model fitting. Similarly, da Silva, Migon, and Correia (2011) proposed a state space Bayesian model for a time series of proportions, extended in da Silva and Rodrigues (2015) to Dirichlet models with a static scale parameter. Zheng and Chen (2017) proposed a frequentist Dirichlet ARMA time series. Much of the prior work on modeling compositional data and compositional time series transforms $\mathbf{y}_t$ from the original J-dimensional simplex to a Euclidean space where the data are now modeled as normal (Aitchison, 1982; AL-Dhurafi, Masseran, & Zamzuri, 2018; Barcelo-Vidal, Aguilar, & Martín-Fernández, 2011; Cargnoni, Müller, & West, 1997; Koehler, Snyder, Ord, Beaumont, et al., 2010; Kynčlová, Filzmoser, & Hron, 2015; Mills, 2010; Ravishanker, Dey, & Iyengar, 2001; Silva & Smith, 2001; Snyder, Ord, Koehler, McLaren, & Beaumont, 2017).

It would seem preferable not to transform the raw data before modeling and instead model the data $\mathbf{y}_t$ directly as Dirichlet distributed. Thus we propose a new class of Bayesian Dirichlet auto-regressive moving average models (B-DARMA) for compositional time series. We model the data as Dirichlet distributed, given the mean and scale, and then transform the J-dimensional mean parameter vector to a $J-1$-dimensional vector. The distributional parameters are then modeled with a vector auto-regressive moving average structure. We also model the Dirichlet scale parameter as a log linear function of time-varying predictors.

We give a general framework and present submodels motivated by our Airbnb lead time data. The B-DARMA model can be applied to data sets with large compositional vectors or few observations. It offers efficiency gains through the choice of priors and/or submodels and provides sensible forecasts on Airbnb data.

We consider normal and horseshoe priors for the VAR and VMA coefficients, and for regression coefficients. Normal priors have long been a default for coefficients, while the horseshoe is a newer choice which allows for varying amounts of shrinkage (Bańbura et al., 2010; Carvalho, Polson, & Scott, 2009, 2010; Huber & Feldkircher, 2019; Kastner & Huber, 2020) depending on the magnitude of the coefficients.

The next section presents the B-DARMA model. Section 3 presents simulation studies comparing B-DARMA to both a frequentist DARMA data model and a transformed data normal VARMA model. Section 4 presents our analysis of the Airbnb data. The paper closes with a short discussion.

2. A Bayesian Dirichlet auto-regressive moving average model

2.1. Data model

We observe a J-component multivariate compositional time series $\mathbf{y}_t = (y_{t1}, \ldots, y_{tJ})'$, observed at consecutive integer valued times $t = 1$ up to the most recent time $t = T$, where $0 < y_{tj} < 1$, and $\mathbf{1}'\mathbf{y}_t = 1$ where $\mathbf{1}$ is a J-vector of ones. We model $\mathbf{y}_t$ as Dirichlet with mean vector $\boldsymbol{\mu}_t = (\mu_{t1}, \ldots, \mu_{tJ})'$, with $0 < \mu_{tj} < 1$, $\mathbf{1}'\boldsymbol{\mu}_t = 1$, and scale parameter $\phi_t > 0$:

$$\mathbf{y}_t|\boldsymbol{\mu}_t, \phi_t \sim \text{Dirichlet}(\phi_t\boldsymbol{\mu}_t), \tag{1}$$

with density $f(\mathbf{y}_t|\boldsymbol{\mu}_t, \phi_t) \propto \prod_{j=1}^{J} y_{tj}^{\phi_t\mu_{tj}-1}$.

We model $\boldsymbol{\mu}_t$ as a function of prior observations $\mathbf{y}_1, \ldots, \mathbf{y}_{t-1}$, prior means $\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_{t-1}$, and known covariates $\boldsymbol{x}_t$ in a generalized linear model framework. As $\boldsymbol{\mu}_t$ and $\mathbf{y}$ are constrained, we model $\boldsymbol{\mu}_t$ after reducing the dimension using the additive log ratio (alr) link:

$$\boldsymbol{\eta}_t \equiv \text{alr}(\boldsymbol{\mu}_t) = \left(\log\left(\frac{\mu_{t1}}{\mu_{tj^*}}\right), \ldots, \log\left(\frac{\mu_{tJ}}{\mu_{tj^*}}\right)\right) \tag{2}$$

where j^* is a chosen reference component $1 \le j^* \le J$, and the element of $\boldsymbol{\eta}_t$ that would correspond to the j^*-th element $\log(\mu_{j^*}/\mu_{j^*}) = 0$ is omitted. The linear predictor $\boldsymbol{\eta}_t$ is a $J-1$-vector taking values in $\mathbb{R}^{J-1}$. Given $\boldsymbol{\eta}_t$, $\boldsymbol{\mu}_t$ is defined by the inverse of Eq. (2), where $\mu_{tj} = \exp(\eta_{tj})/(1 + \sum_{j=1}^{J-1}\exp(\eta_{tj}))$ for $j = 1, \ldots, J$, $j \neq j^*$, and for $j = j*$, $\eta_{tj*} = 1/(1 + \sum_{j=1}^{J-1}\exp(\eta_{tj}))$.

We model $\boldsymbol{\eta}_t$ as a vector auto-regressive moving average (VARMA) process:

$$\boldsymbol{\eta}_t = \sum_{p=1}^{P}\boldsymbol{A}_p(\text{alr}(\mathbf{y}_{t-p}) - \boldsymbol{X}_{t-p}\boldsymbol{\beta}) + \sum_{q=1}^{Q}\boldsymbol{B}_q(\text{alr}(\mathbf{y}_{t-q}) - \boldsymbol{\eta}_{t-q}) + \boldsymbol{X}_t\boldsymbol{\beta} \tag{3}$$

for $t = m+1, \ldots, T$, where $m = \max(P, Q)$, $\boldsymbol{A}_p$ and $\boldsymbol{B}_q$ are $(J-1)\times(J-1)$ coefficient matrices of the respective vector auto-regressive (VAR) and vector moving average (VMA) terms, $\boldsymbol{X}_t$ is a known $(J-1)\times r_\beta$ matrix of deterministic covariates, including an intercept and including seasonal variables and trend as needed, and $\boldsymbol{\beta}$ is an $r_\beta \times 1$ vector of regression coefficients. The form of an intercept in $\boldsymbol{X}_t$ is the $(J-1)\times(J-1)$ identity matrix I_{J-1} as $J-1$ columns in $\boldsymbol{X}_t$. Given an $r_0 \times 1$ vector of covariates $\boldsymbol{x}_t$ for day t, the simplest form of $\boldsymbol{X}_t$ is $\boldsymbol{X}_t = I_{J-1} \otimes \boldsymbol{x}_t$ and $r_\beta = (J-1) * r_0$. The additive log ratio in (3) is a multivariate logit link and leads to elements of matrices $\boldsymbol{A}_p$ and $\boldsymbol{B}_q$ and vector $\boldsymbol{\beta}$ that are log odds ratios. Given $\boldsymbol{\eta}_t$, $\boldsymbol{\mu}_t$ gives the expected allocations of fees to components.

Scale parameter ϕ_t is modeled with a log link as a function of an r_γ-vector of covariates $\boldsymbol{z}_t$:

$$\phi_t = \exp(\boldsymbol{z}_t \boldsymbol{\gamma}), \tag{4}$$

where $\boldsymbol{\gamma}$ is an r_γ-vector of coefficients. In the situation of no covariates for ϕ_t, $\log \phi_t = \gamma$ for all t.

Define the consecutive observations $\mathbf{y}_{a:b} = (\mathbf{y}_a, \ldots, \mathbf{y}_b)'$ for positive integers $a < b$. To be well defined, linear predictor (3) requires having m previous observations $\mathbf{y}_{(t-m):(t-1)}$, and corresponding linear predictors $\boldsymbol{\eta}_{t-m}, \ldots, \boldsymbol{\eta}_{t-1}$. In computing posteriors, we condition on the first m observations $\mathbf{y}_{1:m}$ which then do not contribute to the likelihood. For the corresponding first m linear predictors, on the right-hand side of (3), we set $\boldsymbol{\eta}_1, \ldots, \boldsymbol{\eta}_m$ equal to $\text{alr}(\mathbf{y}_1), \ldots, \text{alr}(\mathbf{y}_m)$, which effectively omits the VMA terms $\boldsymbol{B}_l(\text{alr}(\mathbf{y}_{t-l}))$ from (3) when $t - l \leq m$. In contrast, in (3) the VAR terms and $\boldsymbol{X}_t\boldsymbol{\beta}$ are well defined for $t = 1, \ldots, m$.

Define the C-vector $\boldsymbol{\theta}$ of all unknown parameters $\boldsymbol{\theta} = (\boldsymbol{A}_{prs}, \boldsymbol{B}_{qrs}, \boldsymbol{\beta}', \boldsymbol{\gamma}')'$, where $r, s = 1, \ldots, J-1$ index all elements of matrices $\boldsymbol{A}_p$ and $\boldsymbol{B}_q$, $p = 1, \ldots, P$, $q = 1, \ldots, Q$ and $C = (P+Q)*(J-1)^2 + r_\beta + r_\gamma$. Prior beliefs $p(\boldsymbol{\theta})$ about $\boldsymbol{\theta}$ are updated by Bayes' theorem to give the posterior

$$p(\boldsymbol{\theta}|\mathbf{y}_{1:T}) = \frac{p(\boldsymbol{\theta})p(\mathbf{y}_{(m+1):T}|\boldsymbol{\theta}, \mathbf{y}_{1:m})}{p(\mathbf{y}_{(m+1):T}|\mathbf{y}_{1:m})},$$

where $p(\mathbf{y}_{(m+1):T}|\boldsymbol{\theta}, \mathbf{y}_{1:m}) = \prod_{t=m+1}^{T} p(\mathbf{y}_t|\boldsymbol{\theta}, \mathbf{y}_{(t-m):(t-1)})$, $p(\mathbf{y}_t|\boldsymbol{\theta}, \mathbf{y}_{(t-m):(t-1)})$ is the density of the Dirichlet in (1), and the normalizing constant $p(\mathbf{y}_{(m+1):T}|\mathbf{y}_{1:m}) = \int p(\boldsymbol{\theta})p(\mathbf{y}_{(m+1):T}|\boldsymbol{\theta}, \mathbf{y}_{1:m})d\boldsymbol{\theta}$. We wish to forecast the next S observations $\mathbf{y}_{(T+1):(T+S)}$. These have a joint predictive distribution:

$$p(\mathbf{y}_{(T+1):(T+S)}|\mathbf{y}_{1:T}) = \int_{\boldsymbol{\theta}} p(\mathbf{y}_{(T+1):(T+S)}|\boldsymbol{\theta})p(\boldsymbol{\theta}|\mathbf{y}_{1:T})d\boldsymbol{\theta}.$$

The joint predictive distribution $p(\mathbf{y}_{(T+1):(T+S)}|\mathbf{y}_{1:T})$ can be summarized for example by the mean or median against time $t \in (T+1) : (T+S)$ to communicate results to business managers.

Our data model can be viewed as a Bayesian multivariate Dirichlet extension of the generalized ARMA model by Benjamin et al. (2003). Similarly, Zheng and Chen (2017) propose a DARMA data model whose link function in (2) does not have an analytical inverse and so needs to be approximated numerically. They view our data model as an approximation to theirs, noting that the resulting noise sequence from having the analytical inverse is not a martingale difference sequence (MDS). This lack of an MDS complicates the investigation of the probabilistic properties of the series and the asymptotic behavior of their estimators.

From a Bayesian perspective, the need for an MDS for making inferences is circumvented. Bayesian inference is based on the posterior distribution of the parameters, which combines the likelihood function (dependent on the data) and the prior distribution (representing our prior beliefs about the parameters). This allows us to make inferences regardless of whether the data form an MDS.

Furthermore, the concept of a "noise sequence" or residuals is somewhat different in the context of generalized linear models (GLMs) compared to location-scale models like linear regression. In GLMs, residuals are not explicitly defined, and the noise sequence discussed in the context of Zheng and Chen's model is an approximation. In our Bayesian approach, we do not need to define or consider the noise sequence at all, which simplifies the model and the analysis.

2.2. Choice of link function

The link function in (2) and (3) can be replaced with other common simplex transformations, such as the centered log ratio (CLR):

$$\text{clr}(\mathbf{y}) = \left[\ln\left(\frac{y_1}{g(\mathbf{y})}\right), \ln\left(\frac{y_2}{g(\mathbf{y})}\right), \ldots, \ln\left(\frac{y_J}{g(\mathbf{y})}\right)\right]$$

where $g(\mathbf{y}) = \left(y_1 \times y_2 \times \cdots \times y_J\right)^{1/J}$. Alternatively, the isometric log ratio (ILR) transformation can be used, with the jth component

$$y_j = \sqrt{\frac{r_j}{r_j+1}} \log\left(\frac{g_j(\mathbf{y})}{\left(\prod_{i\in H_j} y_i\right)^{1/r_j}}\right)$$

where $g_j(\mathbf{y})$ is the geometric mean of a subset S_j of $\mathbf{y}$, H_j is the complement of S_j, and r_j is the number of elements in H_j. The choice of subsets S_j and H_j can vary depending on the specific problem and interpretation requirements. Egozcue, Pawlowsky-Glahn, Mateu-Figueras, and Barcelo-Vidal (2003) showed that the coefficient matrices of the ALR, CLR, and ILR are linear transformations of each other. This means that the DARMA data models with these three link functions are equivalent, provided the same transformation is applied to the priors.

2.3. Model selection

For model selection, we use the approximate calculation of the leave-future-out (LFO) estimate of the expected log pointwise predictive density (ELPD) for each model which measures predictive performance (Bernardo & Smith, 2009; Vehtari & Ojanen, 2012):

$$\text{ELPD}_{\text{LFO}} = \sum_{t=L}^{T-M} \log p(\mathbf{y}_{(t+1):(t+M)}|\mathbf{y}_{1:t})$$

where

$$p(\mathbf{y}_{(t+1):(t+M)}|\mathbf{y}_{1:t}) = \int_{\boldsymbol{\theta}} p(\mathbf{y}_{(t+1):(t+M)}|\mathbf{y}_{1:t}, \boldsymbol{\theta}) p(\boldsymbol{\theta}|\mathbf{y}_{1:t}) d\boldsymbol{\theta}, \tag{5}$$

and where M is the number of step-ahead predictions, and L is a chosen minimum number of observations from the time series needed to make predictions. We use Monte Carlo methods to approximate (5) with S^* random draws from the posterior and estimate $p(\mathbf{y}_{(t+1):(t+M)}|\mathbf{y}_{1:t})$ as

$$p(\mathbf{y}_{(t+1):(t+M)}|\mathbf{y}_{1:t}) \approx \frac{1}{S^*} \sum_{s=1}^{S^*} p(\mathbf{y}_{(t+1):(t+M)}|\mathbf{y}_{1:t}, \boldsymbol{\theta}_{1:t}^{(s)})$$

for $s = 1, \ldots, S^*$ draws from the posterior distribution $p(\boldsymbol{\theta}|\mathbf{y}_{1:t})$. Calculating the ELPD LFO requires refitting the model for each $t \in (L, \ldots, T - M)$. To get around this, we use approximate M-step-ahead predictions using Pareto smoothed importance sampling (Bürkner, Gabry, & Vehtari, 2020).

2.4. Priors

A useful vague prior for the individual coefficients in $\boldsymbol{\theta}$ is a proper independent normal $N(V_0, V)$ with varying V_0 and V depending on prior beliefs about the coefficients. We take $V_0 = 0.4, 0.1$ in our data analysis when we believe a priori that the coefficients will be positive and $V_0 = 0$ when we are unsure. For elements a_{prs} and b_{qrs} of $\boldsymbol{A}_p$ and $\boldsymbol{B}_q$ matrices, we take $V = 0.5^2$ as we expect those elements to be between $[-1, 1]$. For elements of $\boldsymbol{\beta}$, we might for example let V vary with the standard deviation of the covariate.

It may be thought that many elements of $\boldsymbol{\theta}$ might be at or near zero. In this case, rather than a normal prior, we might consider a shrinkage prior like that proposed by Carvalho et al. (2010), who propose a horseshoe prior of the form

$$\theta_c|\tau, \lambda_{c^*} \sim N(0, \tau^2\lambda_{c^*}^2)$$
$$\lambda_{c^*} \sim C^+(0, 1).$$

for some $c^* = 1, \ldots, C^*$, $c = 1, \ldots, C$, and $C^* \leq C$. Each θ_c belongs to one of $c^* = 1, \ldots, C^*$ groups, and each group gets its own shrinkage parameter λ_{c^*} applied to it. We apply this with component-specific shrinkage parameters in our data analysis.

3. Simulation study

In two simulation studies, we study the B-DARMA model's capacity to accurately retrieve true parameter values, a critical aspect for reliable forecasting. By comparing B-DARMA with established models on a number of metrics, we provide a robust assessment of both its estimation and forecasting potential. Each simulation study has $L = 400$ data sets indexed by $l = 1, \ldots, L$ with $T = 540$ observations $\mathbf{y}_t^{(l)}$, $t = 1, \ldots, T$ with $J = 3$ components, $j^* = 3$ as the reference component for all models, and $\mathbf{Y}^l = (\mathbf{y}_1^{(l)}, \ldots, \mathbf{y}_T^{(l)})$. We split the 540 observations into a training set of the first 500 observations and a test set of the last 40 observations, the holdout data, which the model is not trained on and which are used to show the models' forecasting performance.

In both studies, we compare B-DARMA to a non-Bayesian DARMA data model and to a non-Bayesian transformed-data normal VARMA (tVARMA) model that transforms $\mathbf{y}_t^{(l)}$ to $\text{alr}(\mathbf{y}_t^{(l)}) \in \mathbb{R}^{J-1}$ and models $\text{alr}(\mathbf{y}_t) \sim N_{J-1}(\boldsymbol{\eta}_t, \boldsymbol{\Sigma})$, where Σ is an unknown $J - 1 \times J - 1$ positive semi-definite matrix, and $\boldsymbol{\eta}_t$ is defined in (3). We use the VARMA function from the MTS package (Tsay, Wood, & Lachmann, 2022) in R (R. Core Team, 2022) to fit the tVARMA models with maximum likelihood. For the non-Bayesian DARMA data model, the BFGS algorithm as implemented in optim in R is used for optimization. To compute the parameters' standard errors, the negative inverse Hessian at the mode is calculated.

The data generating model (DGM) is a DARMA model in simulation 1 and a tVARMA model in simulation 2. We set $p = 1$, $q = 1$ for a DARMA(1, 1) (simulation 1) or tVARMA(1, 1) (simulation 2) data generating model.

To keep the parameterization of the B-DARMA(P, Q) model consistent with the parameterization of the MTS package, we remove $-\boldsymbol{X}_{t-p}\boldsymbol{\beta}$, $p = 1, \ldots, P$ from the VAR term in (3) and set $\boldsymbol{\eta}_t = \sum_{p=1}^{P} \boldsymbol{A}_p(\text{alr}(\mathbf{y}_{t-p})) + \sum_{q=1}^{Q} \boldsymbol{B}_q(\text{alr}(\mathbf{y}_{t-q}) - \boldsymbol{\eta}_{t-q}) + \boldsymbol{X}_t\boldsymbol{\beta}^*$, where $\boldsymbol{\beta}^* = \boldsymbol{\beta} - \sum_{p=1}^{P} \boldsymbol{A}_p\boldsymbol{\beta}$, and we drop the asterisk on $\boldsymbol{\beta}$ for the remainder of this section.

All B-DARMA models are fit with STAN (Stan Development Team, 2022) in R. We run four chains with 2000 iterations each, with a warm up of 1000 iterations for a posterior sample of 4000. Initial values are selected randomly from the interval $[-1, 1]$.

In all simulations, the covariate matrix $\boldsymbol{X}_t = \boldsymbol{I}_2$, the 2×2 identity matrix. Unknown coefficient matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ have dimension 2×2. The shared parameters of interest for the DARMA, B-DARMA, and tVARMA are $\boldsymbol{B} = (b_{rs})$, and $\boldsymbol{A}_p = (a_{prs})$, $r, s = 1, 2$, $p = 1$ and $\boldsymbol{\beta} = (\beta_1, \beta_2)'$. We use the out-of-sample forecast root mean squared error, FRMSE_j, and forecast mean absolute error, FMAE_j, for each component j as a measure of forecasting performance:

$$\text{FRMSE}_j = \left(\left(\frac{1}{400}\right) \left(\frac{1}{40}\right) \sum_{l=1}^{400} \sum_{t=501}^{540} (y_{tj}^l - \bar{\mu}_{tj}^l)^2 \right)^{\frac{1}{2}}$$

$$\text{FMAE}_j = \left(\frac{1}{400}\right) \left(\frac{1}{40}\right) \sum_{l=1}^{400} \sum_{t=501}^{540} |y_{tj}^l - \bar{\mu}_{tj}^l|$$

where $\bar{\mu}_{tj}^l$ is the posterior mean of μ_{tj} or the maximum likelihood estimate in the lth data set. The tVARMA model has an additional unknown covariance matrix $\boldsymbol{\Sigma} = (\Sigma_{rs})$, $r, s = 1, 2$, a function of three unknown parameters, two standard deviations $\sigma_1 = \Sigma_{11}^{1/2}$, $\sigma_2 = \Sigma_{22}^{1/2}$, and correlation $\rho = \Sigma_{12}/(\sigma_1 * \sigma_2)$, while the B-DARMA and DARMA have a single unknown scale parameter ϕ. For the tVARMA data generating simulations, we set $\sigma_1 = \sigma_2 = .05$, $\rho = .30$, and for DARMA generating models, we set $\phi = 1000$.

Priors in simulations 1 and 2 for the B-DARMA(1, 1) are independent $N(0, .5^2)$ for all coefficients in β, A, B, and a

priori $\gamma = \log\phi \sim \text{Gamma}(25/7, 5/7)$ with mean 5 and variance 7.

For each parameter, generically θ, with a true value θ_{true} in simulation l, for Bayesian models we take the posterior mean $\bar{\theta}^l$ as the point estimate. For the 95% credible interval (CI) $(\theta^l_{\text{low}}, \theta^l_{\text{upp}})$ we take the endpoints to be 2.5% and 97.5% quantiles of the posterior. For the tVARMA and DARMA models, we use the maximum likelihood estimates (MLE) as the parameter estimate, and 95% confidence intervals (CIs) calculated as the maximum likelihood estimate plus or minus 1.96 standard errors.

For each parameter θ in turn, for the B-DARMA, tVARMA, and DARMA models, we assess the bias, root mean squared error (RMSE), length of the 95% CI (CIL), and the fraction of simulations % IN where θ_{true} falls within the 95% CI:

$$\text{bias} = \frac{1}{400}\sum_{l=1}^{400}(\bar{\theta}^l - \theta_{\text{true}})$$

$$\text{RMSE} = \left(\frac{1}{400}\sum_{l=1}^{400}(\bar{\theta}^l - \theta_{\text{true}})^2\right)^{1/2}$$

$$\text{CIL} = \frac{1}{400}\sum_{l=1}^{400}\left(\theta^l_{\text{upp}} - \theta^l_{\text{low}}\right)$$

$$\%\ \text{IN} = \frac{1}{400}\sum_{l=1}^{400}\mathbf{1}\{\theta^l_{\text{low}} \le \theta_{\text{true}} \le \theta^l_{\text{upp}}\}$$

for the B-DARMA models and replace $\bar{\theta}^l$ with the MLE for the tVARMA and DARMA models.

For all simulations, elements of $\boldsymbol{\beta}, \boldsymbol{A}, \boldsymbol{B}$ are set to be

$$\beta_1 = -0.07, \beta_2 = 0.10,$$
$$a_{11} = 0.95, a_{12} = -0.18, a_{21} = 0.3, a_{22} = 0.95,$$
$$b_{11} = 0.65, b_{12} = 0.15, b_{21} = 0.2, b_{22} = 0.65.$$

3.1. Comparing B-DARMA, DARMA, and tVARMA models in simulations 1 and 2

Supplementary Tables 4 and 5 provide summarized parameter recovery results, while Table 1 gives the FRMSE_j and FMAE_j for each component. When the data generating model is a DARMA(1,1), B-DARMA consistently outperforms tVARMA, with the B-DARMA(1,1) yielding an RMSE averaging at 40% smaller, especially for the $\boldsymbol{B}$ matrix coefficients. The B-DARMA's performance aligns with that of the frequentist DARMA, each excelling in half of the 10 considered coefficients. Both DARMA models showcase coverage superior to the tVARMA.

For a tVARMA(1,1) data generating model, the tVARMA generally exhibits the smallest RMSE for all coefficients, barring b_{11}. The difference in RMSE is marginal, averaging at 2%. The B-DARMA and frequentist DARMA compare similarly for $\boldsymbol{\beta}$ and $\boldsymbol{A}$, but the DARMA performs worse for $\boldsymbol{B}$. B-DARMA has better coverage than both the tVARMA and frequentist DARMA for most coefficients, excepting β_2, b_{21}, and b_{22}.

The out-of-sample prediction results are summarized in Table 1. When the DGM is a DARMA(1,1), the B-DARMA outperforms both the frequentist DARMA model and the tVARMA in FRMSE and FMAE across all components, most notably for y_3. When the DGM is a tVARMA, for y_1 and y_2 the B-DARMA performs comparably to the tVARMA and outperforms the DARMA. For y_3, the B-DARMA significantly outperforms both the tVARMA and DARMA.

4. Airbnb Lead time data analysis

The Airbnb lead time data, $\mathbf{y}_t$, are compositional time series for a specific single large market where each component is the proportion of fees booked on day t that are to be recognized in 11 consecutive 30-day windows and one last consecutive 35-day window to cover 365 days of lead times.

Figs. 1 and 2 plot the Airbnb lead time data from 01/01/2015 to 01/31/2019 and 03/26/2015 to 05/14/2015, respectively. There is a distinct repeated yearly shape in Fig. 1 and clear weekly seasonality in Fig. 2. Fig. 1 shows a gradual increase/decrease in component sizes, most notably a decrease for the first window of $[0, 29)$ days. The levels are driven by the attractiveness of certain travel periods, with guests booking earlier and paying more for peak periods like summer and the December holiday season. The weekly variation shows a pronounced contrast between weekdays and weekends. Thus, we want to include weekly and yearly seasonal variables and a trend variable in the predictors $\boldsymbol{X}_t$ in (3). We model each component with its own linear trend and use Fourier terms for our seasonal variables, pairs of $\left(\sin\frac{2k\pi t}{w_{\text{season}}}, \cos\frac{2k\pi t}{w_{\text{season}}}\right)$ for $k = 1, \ldots, K_{\text{season}} \le \frac{w_{\text{season}}}{2}$, where we take $w_{\text{season}} = w_{\text{week}} = 7$ for weekly seasonality and $w_{\text{season}} = w_{\text{year}} = 365.25$ for yearly seasonality. The orthogonality of the Fourier terms helps with convergence, which is why we prefer it to other seasonal representations.

We train the model on data from 01/01/2015 to 1/31/2019, choose a forecast window of 365, and use 02/01/2019 to 01/31/2020 as the test set. All B-DARMA models are fit with STAN using the R interface, where we run four chains with 3000 iterations, each with a warm up of 1500 iterations, for a total of 6000 posterior samples. Initial values are selected randomly from the interval $[-1, 1]$.

4.1. Model

We use the approximate LFO ELPD of candidate models to first decide on the number of yearly Fourier terms, K_{year}, given initial choices of P and Q. Then, given the number of Fourier terms, we decide on the orders P and Q. We fix $K_{\text{season}} = K_{\text{week}} = 3$, the pairs of Fourier terms for modeling weekly seasonality, for all models. We use an intercept and the same seasonal variables and linear trend for ϕ_t. To define LFO ELPD, we take the step-ahead predictions to be $M = 1$, the minimum number of observations from the time series to make predictions to be $L = 365$, and the threshold for the Pareto estimates to be .7 (Bürkner et al., 2020; Vehtari, Simpson, Gelman, Yao, & Gabry, 2015), which resulted in needing to refit models at most 12 times.

Table 1
Simulation study – forecast root mean squared error (FRMSE) and forecast mean absolute error (FMAE) on the test set ($T = 40$). DGM is data generating model. Simulation study 1 is on the left and study 2 is on the right.

	DGM:	DARMA(1,1)			tVARMA(1,1)		
	Model:	B-DARMA	DARMA	tVARMA	B-DARMA	DARMA	tVARMA
FRMSE	y_1	0.1054	0.1091	0.1063	0.0566	0.0794	0.0566
	y_2	0.1342	0.1367	0.1356	0.0748	0.0775	0.0748
	y_3	0.1015	0.1082	0.1470	0.0548	0.0640	0.0787
FMAE	y_1	0.0770	0.0777	0.0800	0.0432	0.0633	0.0432
	y_2	0.1004	0.1009	0.1020	0.0574	0.0652	0.0574
	y_3	0.0766	0.0775	0.1102	0.0422	0.0529	0.0617

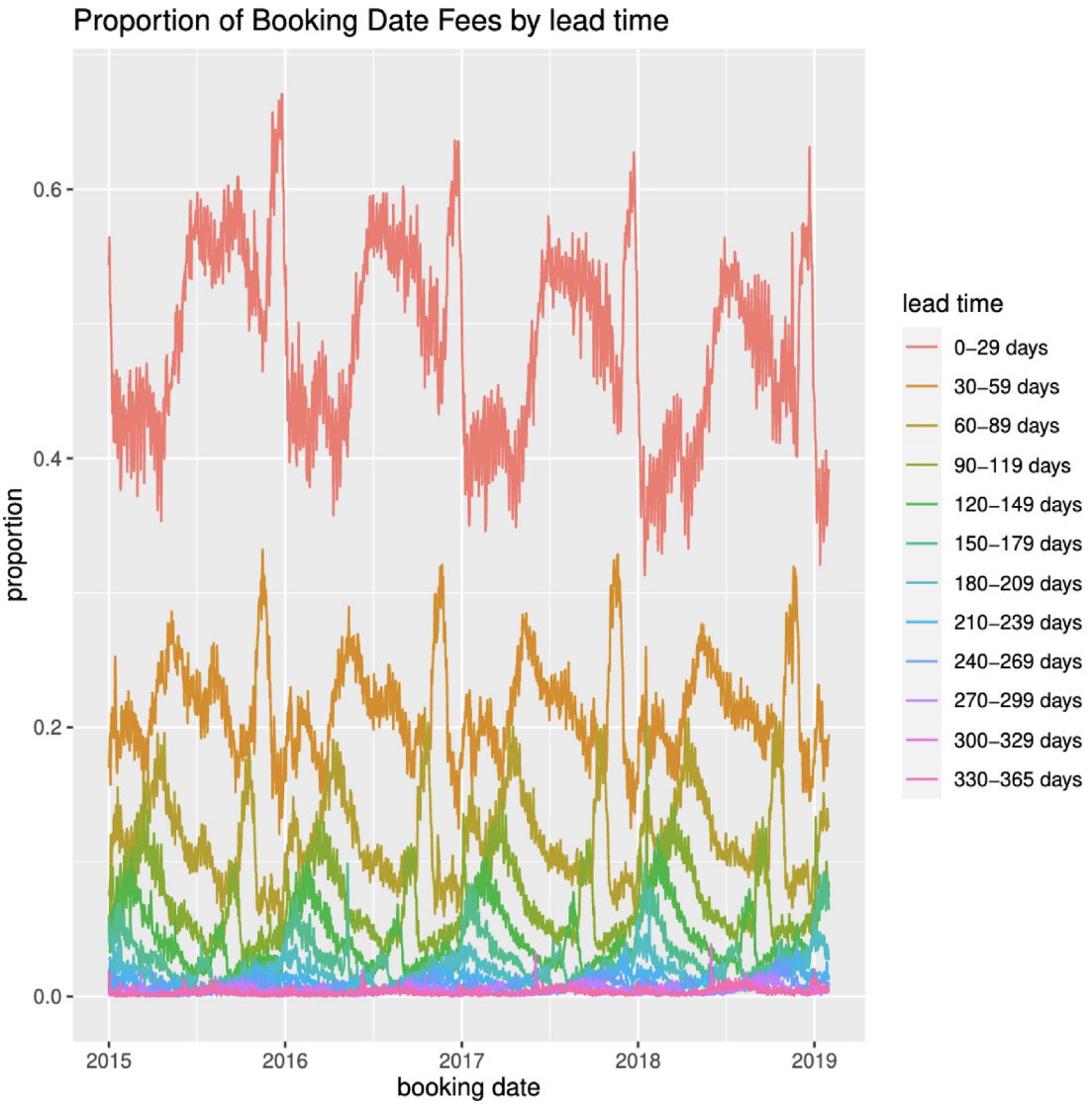


Fig. 1. Airbnb data analysis – proportion of fees by lead time for a single large market from Jan 1, 2015 to Jan 31, 2019.

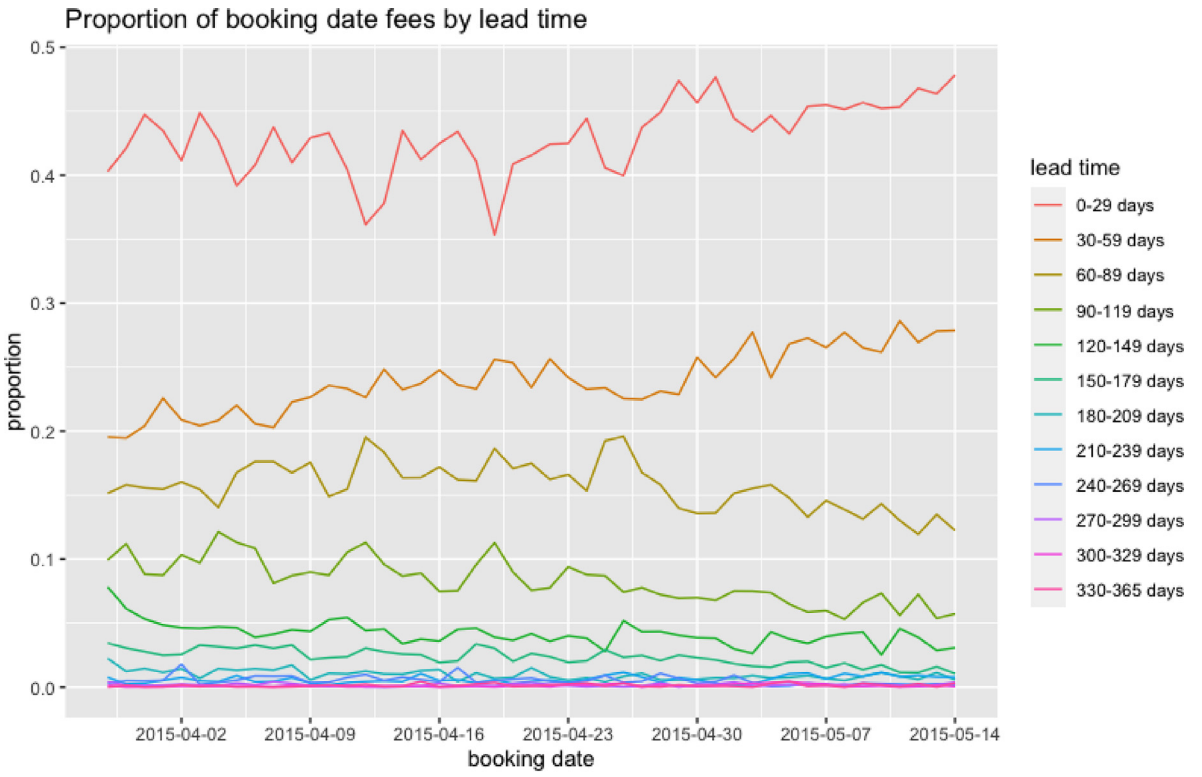


Fig. 2. Airbnb data analysis – proportion of fees for a single large market: weekly seasonal behavior.

For all of the model selection process, we set independent $N(0, .5^2)$ priors for each $(a_{prs}), r, s = 1, \ldots, 11, p = 1, 2$ and $(b_{qrs}), r, s = 1, \ldots, 11, q = 1$, independent $N(0, 1)$ priors on the Fourier coefficients, $N(0, .1)$ priors on the linear trend coefficients, and a $N(0, 4)$ prior on the intercepts. We use these same priors for the seasonal, intercept, and trend coefficients in γ in (4).

We fixed $P = 1$ and $Q = 0$, and let $K_{\text{season}} = K_{\text{year}}$ take on increasing values starting with 3, then sequentially $K_{\text{year}} = 6, 8, 9, 10$, with increasing values of LFO ELPD until 10 showed a worse LFO ELPD. So we stopped and took $K_{\text{year}} = 9$ (Table 2). We fixed $K_{\text{year}} = 9$ and similarly took $(P, Q) = (1, 0)$ first and then $(P, Q) = (1, 1)$ and $(P, Q) = (2, 0)$, with $(1, 0)$ performing best (Table 2).

We compare four different B-DAR(1) models plus a DAR(1) model and a tVAR(1) model. Model 1 (Horseshoe Full) has a horseshoe ($\tau = 1$) prior on the coefficients in $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$, with a separate shrinkage parameter λ_{c^*} for the elements of $\boldsymbol{\theta}$ that correspond to each η_j or to ϕ.

As we expect the AR a_{rs} elements to diminish in magnitude as the time difference $|r - s|$ increases, the other three models have varying prior mean and standard deviations for a_{rs} as functions of $|r - s|$. Model 2 (Normal Full) has an $a_{rr} \sim N(.4, .5^2)$ prior on its diagonal elements $r = 1, \ldots, 11$, an $a_{r(r+1)}, a_{r(r-1)} \sim N(.1, .5^2)$ prior on its two nearest neighbors, $r = 1, \ldots, 10$ and $r = 2, \ldots, 11$, respectively, and an $N(0, .5^2)$ prior on the remaining elements of the $\boldsymbol{A}$ matrix.

In contrast to models 1 and 2, where all parameters a_{rs} in $\boldsymbol{A}$ are allowed to vary, models 3 and 4 fix some of the parameters a_{rs} to 0. Model 3 (Normal Nearest Neighbor) only allows $a_{r(r+1)}, r = 1, \ldots, 10, a_{r(r-1)}, r = 2, \ldots, 11$ and $a_{rr}, r = 1, 2, \ldots, 11$ to vary, and model 4 (Normal Diagonal) only allows $a_{rr}, r = 1, 2, \ldots, 11$ to vary. The remaining elements in model 3 $a_{r(r+k)}$ for $k \geq 2$ or $k \leq -2$ in the $\boldsymbol{A}$ matrix are set equal to 0. For model 4, all off-diagonal elements, a_{rs} $r \neq s$, are set to 0. The non-zero coefficients have the same priors as the Normal Full.

All three Normal models have the same priors on coefficients $\boldsymbol{\gamma}$ and $\boldsymbol{\beta}$: $N(0, 1)$ and $N(0, .1^2)$ for the coefficients of the Fourier and trend terms, respectively. The scale model predictors include the same seasonal and trend terms, and the corresponding priors on the coefficients

Table 2
Airbnb data analysis – leave-future-out expected log pointwise predictive density (LFO ELPD) and the LFO ELPD differences between the best-performing B-DARMA model and candidate B-DARMA models. The first five candidate models fix (P, Q) and vary pairs of Fourier terms for yearly seasonality, and the last two candidate models fix the Fourier terms at nine pairs and vary (P, Q).

(P,Q)	Pairs of Fourier terms	ELPD diff	LFO ELPD
(1,0)	9	0.0	70 135.3
(1,0)	10	−23.3	70 112.0
(1,0)	8	−83.2	70 052.0
(1,0)	6	−594.7	69 540.6
(1,0)	3	−1190.1	68 945.2
(1,1)	9	−101.7	70 033.6
(2,0)	9	−123.7	70 011.5

Table 3
Airbnb data analysis – forecast root mean squared error (FRMSE) and forecast mean absolute error (FMAE) for the 12 components and their total for the test set from Feb 1, 2019 to Jan 31, 2020.

		Normal	Horseshoe	Normal	Normal		
		Full	Full	Nearest neighbor	Diagonal	DAR(1)	tVAR(1)
FRMSE	y_1	0.0211	0.0213	0.0216	0.0225	0.0213	0.0301
	y_2	0.0122	0.0124	0.0129	0.0132	0.0124	0.0145
	y_3	0.0101	0.0101	0.0103	0.0106	0.0101	0.0118
	y_4	0.0094	0.0094	0.0097	0.0099	0.0094	0.0110
	y_5	0.0061	0.0061	0.0062	0.0063	0.0061	0.0076
	y_6	0.0050	0.0050	0.0051	0.0052	0.0050	0.0056
	y_7	0.0037	0.0037	0.0039	0.0040	0.0037	0.0042
	y_8	0.0029	0.0029	0.0030	0.0031	0.0029	0.0033
	y_9	0.0023	0.0023	0.0023	0.0023	0.0023	0.0026
	y_{10}	0.0022	0.0022	0.0023	0.0023	0.0022	0.0023
	y_{11}	0.0025	0.0023	0.0025	0.0025	0.0025	0.0025
	y_{12}	0.0022	0.0022	0.0022	0.0022	0.0022	0.0030
	Total	0.0796	0.0800	0.0821	0.0841	0.0801	0.0983
FMAE	y_1	0.0170	0.0171	0.0174	0.0181	0.0172	0.0247
	y_2	0.0090	0.0094	0.0099	0.0111	0.0096	0.0113
	y_3	0.0076	0.0079	0.0081	0.0083	0.0079	0.0090
	y_4	0.0067	0.0068	0.0070	0.0079	0.0070	0.0085
	y_5	0.0048	0.0049	0.0049	0.0051	0.0049	0.0059
	y_6	0.0039	0.0039	0.0042	0.0042	0.0039	0.0044
	y_7	0.0027	0.0027	0.0027	0.0030	0.0027	0.0032
	y_8	0.0021	0.0021	0.0022	0.0023	0.0022	0.0025
	y_9	0.0018	0.0018	0.0018	0.0019	0.0018	0.0019
	y_{10}	0.0016	0.0016	0.0016	0.0016	0.0016	0.0017
	y_{11}	0.0017	0.0017	0.0017	0.0017	0.0017	0.0018
	y_{12}	0.0016	0.0016	0.0016	0.0017	0.0016	0.0022
	Total	0.0605	0.0615	0.0631	0.0669	0.0621	0.0771

are the same as for $\boldsymbol{\eta}$. The prior on the intercepts in $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ is $N(0, 2^2)$.

Model 5 (DAR(1)) is the frequentist counterpart to our B-DAR(1) Normal Full model with the same seasonal and trend terms for η_j and ϕ. Model 6 (tVAR(1)) is a frequentist transformed VAR model which models $\text{alr}(\mathbf{y}_t) \sim N_{J-1}(\boldsymbol{\eta}_t, \boldsymbol{\Sigma})$. It has the same seasonal and trend variables in $\boldsymbol{\eta}_t$. Both models 5 and 6 are fit in R, model 5 with the BFGS algorithm as implemented in optim, and model 6 with the VARX function in the MTS package.

4.2. Results

In the training set, the Horseshoe Full model has the largest LFO ELPD (Supplementary Table 8). The LFO ELPD for the Normal Full model and the Normal Nearest Neighbor model are close to the Horseshoe Full model with the Normal Diagonal model having a much worse LFO ELPD than the other models.

For the test set, the forecast root mean squared error (FRMSE_j) and the forecast mean absolute error (FMAE_j) for each component is in Table 3 as well as the total forecast root mean squared error and forecast mean absolute error for each model. The Normal Full model performs best on the test set for most components and has the lowest total forecast root mean squared error and smallest total forecast mean absolute error, with the Horseshoe Full model about 1.5% and 0.1% worse, respectively. The differences between the two full models (models 1 and 2) and models 3 and 4 are largest for the larger components as the FRMSEs for the smaller components are closer. For example, for the largest component y_1, the Normal Diagonal model performs over 7% worse than the Normal Full model. The Normal Nearest Neighbor model performs well considering it has 91 fewer parameters than the Normal Full model, performing 3% worse in total than the Normal Full model.

When compared to the frequentist models, the Normal Full models exhibit superior performance in the largest components y_1 and y_2, having a smaller FMAE and a smaller FRMSE_j. For the smaller components y_4 through y_{12}, the differences in FRMSE_j among the models are subtler, although the FMAE values continue to show a more pronounced difference across all components. The DAR(1) does perform better than the subsetted B-DAR(1) models, with a total FRMSE roughly 2.5% smaller than the Normal Nearest Neighbor model and 5% less than the Normal Diagonal model. The tVAR(1) performs significantly worse than all the DAR models, with a total FRMSE about 15% larger than the worst-performing DAR data model, suggesting that the time-invariant nature of $\boldsymbol{\Sigma}$ is inappropriate for the data.

Figs. 3 and 4 plot out-of-sample forecasts ($\bar{\mu}_{tj}$) and residuals of ($\text{alr}(\mathbf{y}_t) - \hat{\boldsymbol{\eta}}_t$) for the Normal Full model. The model captures most of the yearly seasonality, and the residual terms exhibit no consistent positive or negative bias for any of the components, as they are all centered near 0. Much of the remaining residual structure may be explained by market-specific holidays which are not incorporated in the current model.

The estimated yearly and weekly seasonalities for each of the $\boldsymbol{\eta}_j$ (ALR scale) and ϕ (log scale) components in the Normal Full model are detailed in Supplementary Figs. 5 and 6, respectively. Yearly variation is pronounced in

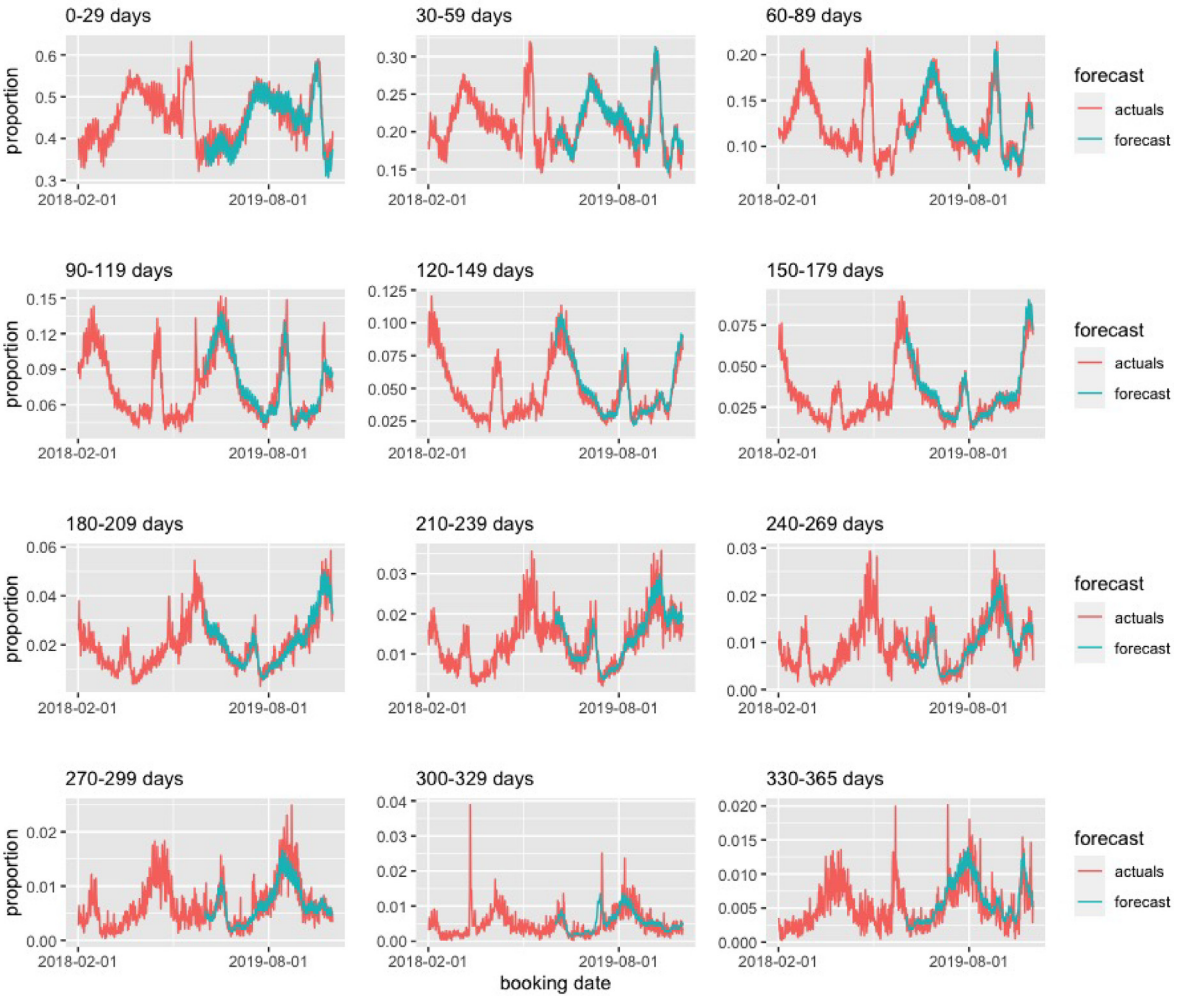


Fig. 3. Airbnb data analysis – Normal Full model: one year of predictions (blue) from Feb 1, 2019 to Jan 31, 2020 and two years of actuals (red) for each of the 12 components. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

larger components, with $\boldsymbol{\eta}_1$ peaking in late December and troughing before the new year, while ϕ exhibits a simpler pattern with two high periods and one low period. Weekly seasonality displays a consistent weekday-versus-weekend behavior across all $\boldsymbol{\eta}_j$ components, showing higher values on weekends. Supplementary Table 9 summarizes the posterior statistics for the intercepts and linear growth rates of $\boldsymbol{\eta}_j$s and $\log(\phi)$, showing a monotonic decrease in the intercepts and larger growth rates for smaller components. The growth rate of ϕ indicates less compositional variability over time. Lastly, the posterior densities for the elements a_{rs} of the $\boldsymbol{A}$ matrix, shown in Supplementary Fig. 7, reveal that larger components have pronounced coefficients on their own lag, a_{rr}, with diminishing magnitude for distant trip dates, while smaller components exhibit more uncertainty but generally maintain a strong positive coefficient on their own lag.

5. Discussion

There are distinct differences in the computational costs of fitting our DARMA model using frequentist or Bayesian methodologies, as well as the tVARMA model using the MTS package in R. Firstly, fitting the tVARMA model using the MTS package was fastest, attributable to its implementation of conditional maximum likelihood estimation with a multivariate Gaussian likelihood. This approach benefits from the mathematical simplicity of estimation under the normal model and its unconstrained optimization problem, resulting in significantly less computational overhead. This is in addition to the built-in performance optimizations found in specialized packages like MTS.

In contrast, the Bayesian DARMA model, implemented using Hamiltonian Monte Carlo (HMC), had higher computational costs. This is to be expected, given that HMC

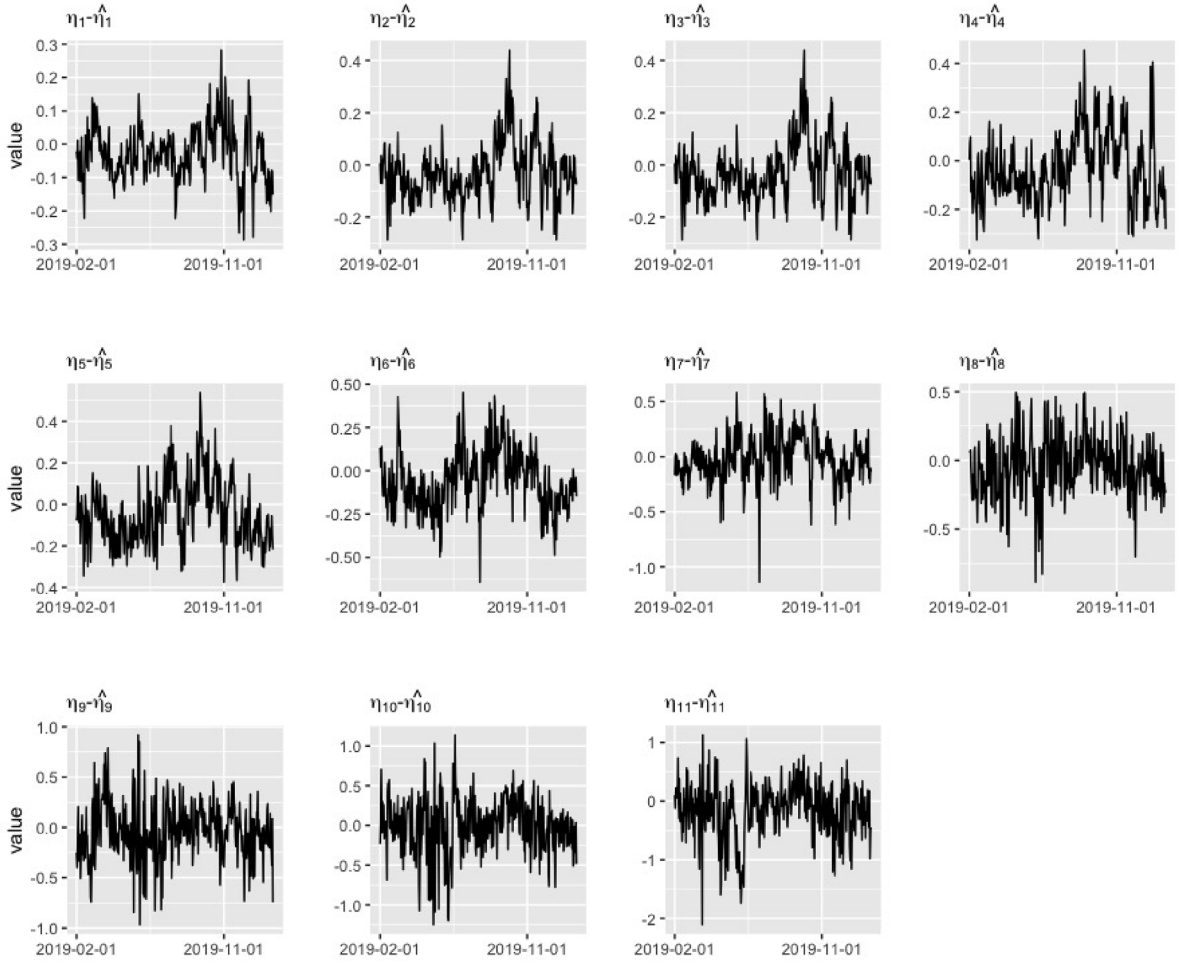


Fig. 4. Airbnb data analysis – Normal Full model residuals on the additive log ratio scale, $\boldsymbol{\eta}_t - \hat{\boldsymbol{\eta}}_t$, for the test set, Feb 1, 2019 to Jan 31, 2020.

requires running many iterations to generate a representative sequence of samples from the posterior distribution. However, the B-DARMA fit faster than the frequentist DARMA. The frequentist DARMA model, fitted using the Broyden–Fletcher–Goldfarb–Shanno (BFGS) optimization method, was the most computationally demanding in our study. The complexity arises from the calculation of the Hessian matrix estimates, the computation of gamma functions in the Dirichlet likelihood, and the constraint of positive parameters. Notably, around 25% of model fitting attempts in the simulation studies initially failed and required regenerating data and refitting the model, underscoring the computational challenges of this approach.

This paper presented a new class of Bayesian compositional time series models assuming a Dirichlet conditional distribution of the observations, with time-varying Dirichlet parameters that are modeled with a VARMA structure. The B-DARMA outperformed the frequentist tVARMA and DARMA when the underlying data generating model was a DARMA, and it did comparably well to the tVARMA and outperforms the DARMA when tVARMA was the data generating model in the simulation studies. By choice of prior and model subsets, we can reduce the number of coefficients that need to be estimated and better handle data sets with fewer observations. This class of models effectively models the fee lead time behavior at Airbnb, outperforming the DAR and tVAR with the same covariates, and provides an interpretable, flexible framework.

Further development is possible. Common compositional time series have components with no ordering, while the 12 lead time components in the Airbnb data set are time ordered. This suggests that $\boldsymbol{\theta}$ can be modeled hierarchically. The construction of the nearest-neighbor and diagonal models for $\boldsymbol{A}_1$ are examples of using time ordering in model specification. While the distributional parameters $\boldsymbol{\mu}_t$ and ϕ_t are time varying, the elements of $\boldsymbol{\theta}$ are static and thus the seasonality is constant over time, which may not be appropriate for other data sets. We model ϕ_t with exogenous covariates, but our approach is

flexible enough to allow for it to have a more complex AR structure itself. While not an issue for the Airbnb data set, the model's inability to handle zeroes would need to be addressed for data sets with exact zeroes.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Harrison Katz reports a relationship with Airbnb that includes: employment. Kai Brusch reports a relationship with Airbnb that includes: employment.

Code and data availability

The primary dataset used in our study is not publicly available due to confidentiality constraints. However, the Stan code for the Bayesian Dirichlet Auto-Regressive Moving Average (B-DARMA) model is available for public access. It can be found at our GitHub repository: https://github.com/harrisonekatz/B-DARMA-paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijforecast.2024.01.004.

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